

Theory Answers – Written Exam 2026-01-20

Source: [exam-prep-prompts/sample-exam-day/20260120.pdf](#)

Exercise 1

Exercise 1.1

What they ask: Write the weak formulation of the elliptic problem.

Answer to write:

Assume $f \in L^2(\Omega)$ and take

$$V = H_0^1(\Omega).$$

Find $u \in V$ such that, for every $v \in V$,

$$a(u, v) = F(v),$$

where

$$a(u, v) = \alpha \int_{\Omega} \nabla u \cdot \nabla v \, dx + \gamma \int_{\Omega} uv \, dx, \quad F(v) = \int_{\Omega} f v \, dx.$$

Minimal explanation:

Multiply the PDE by a test function v , integrate on Ω , and integrate by parts the diffusion term. The boundary term is zero because $v = 0$ on $\partial\Omega$. The homogeneous Dirichlet condition is included in the space $H_0^1(\Omega)$.

Risk check:

This answer assumes Ω is bounded/Lipschitz and $f \in L^2(\Omega)$, or more generally $f \in V'$.

Exercise 1.2

What they ask: Prove existence and uniqueness of the weak solution.

Answer to write:

Use the Lax–Milgram theorem on $V = H_0^1(\Omega)$. The form a is continuous:

$$|a(u, v)| \leq C \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}.$$

It is coercive because, using Poincaré inequality,

$$a(v, v) = \alpha \|\nabla v\|_{L^2(\Omega)}^2 + \gamma \|v\|_{L^2(\Omega)}^2 \geq \alpha \|\nabla v\|_{L^2(\Omega)}^2 \geq c \|v\|_{H^1(\Omega)}^2.$$

Also

$$|F(v)| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \leq C \|f\|_{L^2(\Omega)} \|v\|_{H^1(\Omega)},$$

so F is continuous. Therefore there exists a unique $u \in V$.

Minimal explanation:

Lax–Milgram gives existence and uniqueness if the bilinear form is continuous and coercive, and the right-hand side is continuous. Coercivity still works when $\gamma = 0$ because $u = 0$ on the boundary gives Poincaré inequality.

Risk check:

Do not say coercivity comes from $\gamma > 0$, because the text only assumes $\gamma \geq 0$. The safe reason is $\alpha > 0$ plus Poincaré.

Exercise 1.3

What they ask: Give the finite element approximation with degree 2 polynomials, plus stability and error estimates.

Answer to write:

Let $\mathcal{T}_h$ be a mesh of Ω and define the P_2 space

$$V_h = \left\{ v_h \in H_0^1(\Omega) : v_h|_K \in \mathbb{P}_2(K) \text{ for all } K \in \mathcal{T}_h \right\}.$$

Find $u_h \in V_h$ such that

$$a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h.$$

Using a basis $\{\varphi_i\}$, this gives $AU = F$, with

$$A_{ij} = \alpha \int_{\Omega} \nabla \varphi_j \cdot \nabla \varphi_i \, dx + \gamma \int_{\Omega} \varphi_j \varphi_i \, dx, \quad F_i = \int_{\Omega} f \varphi_i \, dx.$$

Stability:

$$\|u_h\|_{H^1(\Omega)} \leq C \|f\|_{L^2(\Omega)}.$$

If $u \in H^3(\Omega)$, then for P_2 elements

$$\|u - u_h\|_{H^1(\Omega)} \leq Ch^2 |u|_{H^3(\Omega)}, \quad \|u - u_h\|_{L^2(\Omega)} \leq Ch^3 |u|_{H^3(\Omega)}.$$

Minimal explanation:

The discrete problem is the same weak formulation restricted to a finite-dimensional subspace. Stability follows from coercivity. The error estimate is Cea lemma plus the interpolation estimate for quadratic finite elements.

Risk check:

The L^2 estimate needs enough regularity/elliptic regularity. If the solution is less smooth, the order can be lower.

Exercise 1.4

What they ask: Write one Dirichlet–Neumann iteration on two subdomains and state convergence.

Answer to write:

Let $\Gamma = \partial\Omega_1 \cap \partial\Omega_2$ be the interface. Given an interface guess g^k on Γ :

1. Solve on Ω_1

$$-\operatorname{div}(\alpha \nabla u_1^{k+1}) + \gamma u_1^{k+1} = f,$$

with $u_1^{k+1} = 0$ on $\partial\Omega_1 \cap \partial\Omega$ and

$$u_1^{k+1} = g^k \quad \text{on } \Gamma.$$

2. Solve on Ω_2

$$-\operatorname{div}(\alpha \nabla u_2^{k+1}) + \gamma u_2^{k+1} = f,$$

with $u_2^{k+1} = 0$ on $\partial\Omega_2 \cap \partial\Omega$ and Neumann interface condition

$$\alpha \frac{\partial u_2^{k+1}}{\partial n_2} = -\alpha \frac{\partial u_1^{k+1}}{\partial n_1} \quad \text{on } \Gamma.$$

3. Update the interface value with relaxation:

$$g^{k+1} = (1 - \lambda)g^k + \lambda u_2^{k+1}|_{\Gamma}, \quad 0 < \lambda \leq 1.$$

The method is a fixed-point iteration on the interface. In general, convergence is not automatic for $\lambda = 1$; for elliptic coercive problems it is convergent with a suitable relaxation parameter, often chosen smaller than 1.

Minimal explanation:

One subdomain receives a Dirichlet value on the interface. The other receives the corresponding Neumann flux. The final update tries to make the two traces equal on the interface.

Risk check:

The sign of the Neumann condition depends on the choice of outward normals. The safe condition is flux continuity:

$$\alpha \partial_{n_1} u_1 + \alpha \partial_{n_2} u_2 = 0 \quad \text{on } \Gamma.$$

Exercise 1.5

What they ask: Implement the Dirichlet–Neumann solver in deal.II.

Answer to write:

Da gestire nella parte pratica: codice deal.II del solver con metodo Dirichlet–Neumann rilassato.

Minimal explanation:

This point asks for implementation and source code upload, not for a theory answer.

Risk check:

Reference the uploaded source file on the paper.

Exercise 1.6

What they ask: Run 15 Dirichlet–Neumann iterations and upload plots for two relaxation values.

Answer to write:

Da gestire nella parte pratica: produrre i plot per $\lambda = 1.0$ e $\lambda = 0.25$.

Minimal explanation:

This point asks for numerical plots, not for a written theoretical derivation.

Risk check:

On the paper, reference the plot filenames and specify which plot corresponds to each λ .

Exercise 2

Exercise 2.1

What they ask: Write the weak formulation of the time-dependent Stokes problem.

Answer to write:

Define

$$V = \left\{ v \in [H^1(\Omega)]^2 : v = 0 \text{ on } \Gamma_D \right\}, \quad Q = L^2(\Omega)$$

or $Q = L_0^2(\Omega)$ if a pressure normalization is required. Find

$$u(t) \in V, \quad p(t) \in Q,$$

with $u(0) = 0$, such that for all $v \in V$, $q \in Q$,

$$\begin{aligned} (\partial_t u, v)_\Omega + \mu(\nabla u, \nabla v)_\Omega - (p, \operatorname{div} v)_\Omega &= \int_{\Gamma_N} \psi \cdot v \, ds, \\ (\operatorname{div} u, q)_\Omega &= 0. \end{aligned}$$

Minimal explanation:

The Dirichlet condition is imposed in the velocity space. The Neumann/traction condition becomes the boundary term on Γ_N . The pressure appears through the term $-(p, \operatorname{div} v)$.

Risk check:

If the course defines $b(v, p) = -(p, \operatorname{div} v)$, keep this sign consistently. Some notes may write the incompressibility equation as $b(u, q) = 0$.

Exercise 2.2

What they ask: Give the spatial Taylor–Hood approximation with degree 3 for velocity and degree 2 for pressure.

Answer to write:

Let $\mathcal{T}_h$ be a mesh. Use Taylor–Hood spaces

$$\begin{aligned} V_h &= \left\{ v_h \in V \cap [C^0(\overline{\Omega})]^2 : v_h|_K \in [\mathbb{P}_3(K)]^2 \right\}, \\ Q_h &= \left\{ q_h \in Q \cap C^0(\overline{\Omega}) : q_h|_K \in \mathbb{P}_2(K) \right\}. \end{aligned}$$

Find $u_h(t) \in V_h$, $p_h(t) \in Q_h$, with $u_h(0) = 0$, such that

$$\begin{aligned} (\partial_t u_h, v_h) + \mu(\nabla u_h, \nabla v_h) - (p_h, \operatorname{div} v_h) &= \int_{\Gamma_N} \psi \cdot v_h \, ds, \\ (\operatorname{div} u_h, q_h) &= 0 \end{aligned}$$

for all $v_h \in V_h$, $q_h \in Q_h$.

Minimal explanation:

Taylor–Hood means continuous velocity polynomials of degree one higher than pressure. The pair P_3/P_2 is a standard inf-sup stable pair for Stokes.

Risk check:

If the pressure is required to have zero mean, replace Q_h by $Q_h \cap L_0^2(\Omega)$.

Exercise 2.3

What they ask: Apply backward Euler in time and explain uniqueness at each timestep.

Answer to write:

Let $t_n = n\Delta t$. Given u_h^n , find $u_h^{n+1} \in V_h$, $p_h^{n+1} \in Q_h$ such that

$$\left(\frac{u_h^{n+1} - u_h^n}{\Delta t}, v_h \right) + \mu(\nabla u_h^{n+1}, \nabla v_h) - (p_h^{n+1}, \operatorname{div} v_h) = \int_{\Gamma_N} \psi(t_{n+1}) \cdot v_h \, ds,$$
$$(\operatorname{div} u_h^{n+1}, q_h) = 0.$$

In matrix form:

$$\begin{bmatrix} M/\Delta t + \mu A & B^T \\ B & 0 \end{bmatrix} \begin{bmatrix} U^{n+1} \\ P^{n+1} \end{bmatrix} = \begin{bmatrix} MU^n/\Delta t + F^{n+1} \\ 0 \end{bmatrix}.$$

At every timestep the velocity bilinear form

$$c(u, v) = \frac{1}{\Delta t}(u, v) + \mu(\nabla u, \nabla v)$$

is coercive on V_h . Since the Taylor–Hood pair satisfies the discrete inf-sup condition, the saddle-point problem has a unique solution.

Minimal explanation:

Backward Euler evaluates the unknown velocity and pressure at the new time t_{n+1} . The mass term $M/\Delta t$ helps coercivity. Inf-sup stability controls the pressure.

Risk check:

Pressure uniqueness may require fixing the pressure space, for example by zero mean, depending on the boundary conditions and the convention used in the course.

Exercise 2.4

What they ask: State the expected error estimate in terms of Δt and h .

Answer to write:

For the P_3/P_2 Taylor–Hood pair and backward Euler, assuming the exact solution is smooth enough, the expected orders are

$$\|u(t_n) - u_h^n\|_{H^1(\Omega)} + \|p(t_n) - p_h^n\|_{L^2(\Omega)} \leq C(h^3 + \Delta t),$$

and

$$\|u(t_n) - u_h^n\|_{L^2(\Omega)} \leq C(h^4 + \Delta t).$$

So the method is first order in time, order 3 in space for velocity H^1 and pressure L^2 , and order 4 in space for velocity L^2 .

Minimal explanation:

Backward Euler gives the Δt term. Taylor–Hood P_k/P_{k-1} gives velocity H^1 and pressure L^2 errors of order h^k ; here $k = 3$.

Risk check:

These are expected estimates under regularity assumptions. If Exercise 2.5 uses P_2/P_1 , the corresponding spatial orders become h^2 in velocity H^1 /pressure L^2 and h^3 in velocity L^2 .

Exercise 2.5

What they ask: Implement the time-dependent Stokes solver in deal.II.

Answer to write:

Da gestire nella parte pratica: codice deal.II del solver Stokes instazionario con Taylor–Hood P_2/P_1 .

Minimal explanation:

This point asks for implementation and source code upload, not for a theory answer.

Risk check:

Exercise 2.5 uses P_2/P_1 , while Exercise 2.2 asks theoretically for P_3/P_2 .

Exercise 2.6

What they ask: Compute the solution and upload plots/profiles.

Answer to write:

Da gestire nella parte pratica: plot finale, profilo lungo $y = x$, e plot temporale nel punto richiesto.

Minimal explanation:

This point asks for numerical post-processing and uploaded figures.

Risk check:

On the paper, reference the exact filenames for the final-time plot, the line profile, and the time trace.